

Economic evaluation of Miscanthus production – A review



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ABSTRACT

This paper reviews the peer-reviewed literature dealing with the economics of Miscanthus cultivation, to identify factors influencing the adoption of Miscanthus and to reveal shortcomings in research. In total, 51 studies have been identified for this review. The majority of these publications are recent (i.e. the majority is published after 2009) and concern case studies in Europe (particularly the UK and Ireland) and North America. This review reveals that the economic viability of Miscanthus depends on largely uncertain assumptions especially concerning yields (10–48 t dry matter per ha) and prices (48–134 €/t dry matter) but also concerning the lifespan (10–20 years) and different cost items. A lack of established markets, high establishment costs and uncertainties, arising to a large extent from the long term commitment, are identified as major barriers to Miscanthus adoption. In addition, the level of support for Miscanthus production (e.g. via subsidies) is identified as crucial for Miscanthus profitability, but is found to be highly heterogeneous across countries. Next to diversity in agronomic and economic assumptions, also the assessment criteria and research techniques used to investigate the profitability of Miscanthus differ widely. While the net present value criterion was most frequently used, we identify a lack of approaches that account for risks and uncertainties, which seem to potentially play a critical role in the uptake of Miscanthus by farmers.

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1. Introduction

Problems related to CO₂ emissions and increasing worldwide demand for energy reveal a discrepancy between ecological and economic objectives [1]. Therefore, energy sources with a high degree of sustainability in terms of CO₂ reduction but still ensuring the security of supply are required [2,3]. High yielding perennial energy crops are expected to contribute to developments in this direction by providing CO₂ “neutral” feedstock for energy production [4].

Despite this potentially high demand for renewable energy in the future, supported by various national and transnational climate politics and targets such as a 10% increase in energy use efficiency and a share of 20% of energy supply from “renewables” in the EU up to 2020 [5], the adoption of dedicated perennial biomass crops by farmers remains low [6]. While the contribution of renewable energy from hydro energy plants, wind turbines, solar energy and especially biomass (wood residues, renewable wastes) is remarkable [7], perennial energy crops play a minor role in terms of growing volume [8]. This observation contrasts the finding that perennial energy crops might be interesting for farmers from a financial point of view [9,10] and thus may have a massive potential for renewable energy provision. From a societal perspective, energy production which is less dependent from “stochastic” events like sunshine and wind (which are necessary for solar and wind power generation) should be favored, since constantly available energy supply at any day and night time can be seen as a fundamental pillar of welfare [11]. Hence, it is not surprising that the worldwide share of woody biomass dedicated for energy production amounts to more than 50% of the total harvested woody biomass (> 1.85 of the total 3.6 billion m³ round wood in 2013 [12]) leading to a progressive depletion of natural timber resources [13]. As advantages of biomass for energy purposes are obvious on the one hand and natural resources are in danger on the other hand, viable substitutes for natural grown wood have to be identified. In this context, several perennial energy crops are considered as interesting alternatives, e.g. Switchgrass (*Panicum virgatum*) which has been assessed thoroughly in the USA as a potential high yielding biomass plant [14–16] or the cultivation of fast growing trees in short rotation coppices (src) [6,17,18]. Of particular importance is also Miscanthus because of its unique combination of unpretentiousness regarding management, soil properties and climate with a high annual yield potential over a lifespan of up to 20 years. Furthermore, it has been shown that Miscanthus is one of the most productive bioenergy crops [15,19,20] (e.g. compared to src) converting solar energy into biomass energy up to 30% more efficiently than other crops due to its C4 type of photosynthesis [21], requiring only low nutrient applications [22] while at the same time providing environmental benefits such as sequestration of carbon in the soil, protection against erosion, removal of heavy metals or increased biodiversity [20,23–25]. Thus, based on these and other indicators Miscanthus performs well in comparison to other biomass sources (e.g. [26,27]). Despite this potential relevant role of Miscanthus, little evidence regarding its economic viability has been provided and no systematic review of these aspects has been conducted across different production, economic and political systems. Furthermore, neither systematic information on

drivers and barriers of adoption have been provided, nor has been the methodological assumption underlying the economic assessment of Miscanthus critically reviewed. This article aims to contribute to fill these research gaps by reviewing the existing scientific literature dealing with the economics of Miscanthus production to identify factors influencing the adoption of Miscanthus and to reveal shortcomings in research that require further examination. Such review is potentially relevant for further scientific research dealing with the economic and agronomic aspects of Miscanthus cultivation and especially with the process of decision making of farmers concerning Miscanthus (and comparable biomass crops) adoption. In addition, this review provides valuable information to those who are involved in promoting sustainable energy production such as extension services and politicians. To them, this review will serve as a guideline, leading to a more sophisticated and realistic insight in the economics of Miscanthus production.

The remainder of this article is structured as follows: first, general aspects of Miscanthus cultivation and key characteristics are presented in Section 2. Next, the review procedure is presented in Section 3 containing a description of the examined literature. Subsequently, results concerning costs and benefits of Miscanthus cultivation are presented in Section 4. Shortcomings and implications for future research are discussed in Section 5 followed by a conclusion in Section 6.

2. Background on Miscanthus production

Miscanthus is a large perennial and rhizomatous grass which was imported to Europe from East-Asia as an ornamental plant in the 1930s [22]. While there is a large variety of Miscanthus species, literature assessing the economics of Miscanthus cultivation nearly exclusively deals with *Miscanthus* × *Giganteus* [28] which is a hybrid of *Miscanthus sinensis* and *Miscanthus sacchiflorus* [29]. In this paper it is referred to as Miscanthus. Besides using Miscanthus for energetic uses, this perennial crop also has the potential to serve as feedstock for textiles or construction materials [29,30]. With its C4 photosynthesis process, typical for grasses of arid regions like sugar cane, millet or corn, Miscanthus converts sunlight energy more efficiently into biomass energy than C3 crops and has the ability to cope with dry and hot locations that are exposed to high solar radiation, comparable to cropland which is alternatively dedicated to corn production [28,31]. Nevertheless also in cooler climates Miscanthus has proven high productivity, which qualifies it as potential biomass plant for a central to northern European setting, where a high demand for biomass for energy and other uses is apparent (e.g. [5]).

Miscanthus (*Miscanthus* × *Giganteus*) is considered as a sterile and non-invasive plant. Therefore vegetative propagation of Miscanthus is required [28]. Plantation of rhizomes or pre-cultivated plantlets (micro-propagation) with a density of 1–2 per m² [32,25] takes place in spring with attention to frost since rhizomes are sensitive to temperatures lower than −3 °C [22]. After soil preparation and plantation, which can be carried out mostly with common farm equipment such as plows and a (modified) potato planter [29], an establishment period of 2–4 [33,34] years follows, where only limited biomass yields are

achieved. During this establishment period maintenance work such as weed control and replanting of frost losses is necessary until *Miscanthus* reaches its full competitiveness [29]. Fertilization needs are low compared to conventional crops, but clearly depend on soil properties and nutrient removal from the field. The latter depends in particular on the harvest date [28]. Storing huge amounts of nutrients such as N and K in its rhizomes during winter, a harvest date in spring allows *Miscanthus* to get along with low fertilization since only the withered above-ground parts of the plants are harvested [28]. Additionally, only stems are harvested, while the litter remains on the ground, forming a layer of mulch. Another advantage of a late harvest date is the high share of dry matter (moisture content $\sim 15\%$ [30]) with low mineral content of the harvested material thus leading to reduced costs for transport, storage and a higher (calorific) quality [22]. Harvest can be carried out with conventional farm equipment like self-propelled corn harvesters and balers. Due to the high costs of this machinery, *Miscanthus* harvest is usually outsourced to contractors [9].¹ The yield potential of a fully established 3–4 m tall *Miscanthus* plantation usually reaches from 10 to 20 t dm/ha (tons dry matter per ha) [35] although extremes range from 2.5 to 60 t dm/ha [36,37] depending on local agronomic conditions. The energy content of *Miscanthus* is comparable to wood with around 17 MJ/t² (mega joule per ton) [38,39] potentially enabling *Miscanthus* to act as a substitute for woody biomass.

As a perennial crop, *Miscanthus* needs to be planted only once (apart from selective replanting after the first winter) and afterwards provides annual yields over a lifespan of 10–20 years [36,40]. At the end of its lifespan, when yields are decreasing, the *Miscanthus* plantation is closed using herbicides (e.g. glyphosate) to kill the plants and underground parts of the plant are removed [9].

3. Literature review and data collection

3.1. Literature processing

A comprehensive literature review was conducted to create an overview about the entirety of scientific papers dealing with the economics of *Miscanthus* cultivation. The databases Google Scholar, Web of Science™ and ScienceDirect® were screened for “*Miscanthus*” plus the following keywords: farm level, decision, adoption, cost, return, competitiveness, profit, NPV, benefits, economics, margin, viability, feasibility, real option, biomass and bioenergy. The total number of identified titles (16,353) was screened for economic relevance and if found to be suitable, transferred to Citavi 4.4. The collection of papers was conducted in the beginning of January 2015, so that only papers till 2014 are considered. We focused on peer-reviewed papers published in journals and refrained from dissertations, conference papers (except for 3 papers considered as particularly valuable [41–43]) and other not peer-reviewed articles found during the literature review. Of the initially found 16,353 titles, 96 were found to be relevant in a wider sense. Those pre-selected 96 titles were then scanned thoroughly for their content (the list of these papers is available from the authors upon request). Finally 51 papers were identified as suitable, 45 papers were rejected for different reasons

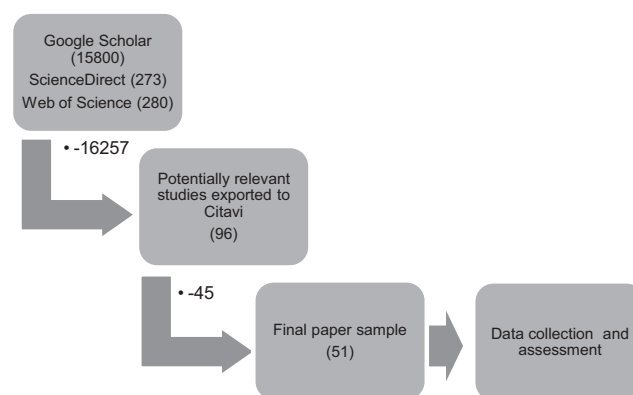


Fig. 1. Illustration of the literature reviewing procedure. The numbers in brackets represent the number of studies on each step of the procedure. The numbers next to the arrows represent the studies excluded during the review process.

such as doublets, *Miscanthus* being only superficially addressed, objectives of papers not being economic but mainly technical aspects of *Miscanthus* cultivation or incomplete sources.³ Other studies have not been considered further because they focused on the use of *Miscanthus* for electricity or fuel generation or on the influences of certain soil qualities, but did not focus on economic aspects (Fig. 1).

The selected papers were surveyed with respect to a wide range of items, concerning (i) the origin of the study and the year of its publication, (ii) the methods used in the investigation, and (iii) assumptions made in the assessment. These comprise details on the production process e.g. whether *Miscanthus* is harvested with a chipper or with a baler, costs, discount rates, the reference system used to compare *Miscanthus* cultivation to other businesses e.g. corn or soybean production, lifespan and harvest time, yields, prices and financial support by subsidization. Our aim was to identify those factors that are most crucial for the profitability of *Miscanthus* and thus are considered to be critical for the farmers' decision regarding the adoption of *Miscanthus*. This causal relation between profitability and adoption of a crop is based on the assumption that farmers will only be willing to change their conventional farming business, if the alternative provides them at least with the same profits or utility⁴ than the current farm business [15,44]. To capture the entire range of possibly decisive factors, the initial data collection process had a wide perspective to avoid that potentially important information were omitted. In this review, we restrict the presentation to factors that have been found to be most important and information, where an amount of data could be retrieved that allows for comparison across studies. For instance, assumptions on the potentially available area of land that could be committed to *Miscanthus* on a national level were not seen as important decision basis for single farmers, while information about assumed lifespans of *Miscanthus* or applied discount rates were seen as highly interesting for adoption decisions. Furthermore, at this stage of the review process, adjustments were made to make the papers originating from different countries comparable. Hence, all prices and costs, stated in different currencies, were transferred to Euro using the respective exchange rates⁵ from February 10th 2015. An overview about the

¹ The fact that demand for this equipment is not within the time-window of maize harvesting is nevertheless assumed to be an advantage for the availability and the price of the contractor.

² Calorific value of *Miscanthus* depends on moisture content, storage conditions and the composition of the feedstock. Thus, precise energy yields have to be calculated for each case individually.

³ This refers to the fact that despite efforts four titles exported to Citavi could not be made available and are therefore not considered.

⁴ Extending the profit framework towards utility allows to also consider aspects of uncertainty, which are of highest relevance for farmers' decision to adopt perennial energy crops (e.g. [18]).

⁵ The respective exchange rates used in our analysis are: 1 US\$ = 0.884€; 1 CAN\$ = 0.707€; 1 £ = 1.345€ (taken from www.oanda.com).

Table 1
Literature summary, overview and description of the reviewed literature.

[illegible]

42.	IT	2014	oil and coconut palm, jatropha, castor bean, sunflower, rapeseed, sugar cane, maize, wheat, sorghum, poplar, cardoon, giant reed, switchgrass, fiber	dc, btl	15	18.00	55.00	10	NPV, CBA	[19]
43.	CZ	2012	src willow, tritiale	dc	10			8.7	NPV	[36]
44.	PL	2012	src poplar	dc	20	14.00	70.30	6.0	NPV, AN, SA	[34]
45.	GE	2012	src poplar	dc	21		89.38		AN, MCS	[70]
46.	NL	1999	src willow	dc, construction	15	20.00				[30]
47.	NL	2010	sugar beet	btl	20			5.5	NPV	[40]
48.	CAN	2013	switchgrass	dc, cf, btl, bedding	15	29.60		5	NPV, BEC	[71]
49.	CAN	2012		dc, cf	20			5	NPV, SA, BEC	[72]
50.	TW	2011	switchgrass	dc, cf, btl	10–20				LCP	[73]
51.	SW	2009	src willow, poplar, eucalyptus, reed canary grass, switch grass, hemp, tritiale, sorghum	dc, btl	21	14.00	79.92	6	NPV, AN	[23]

Explanation of abbreviations: UK = United Kingdom; PL = Poland; HU = Hungary; IT = Italy; LI = Lithuania; IRL = Ireland; US = United States; GR = Greece; F = France; CZ = Czech; GE = Germany; NL = Netherlands; CAN = Canada; TW = Taiwan; SW = Sweden; src = short rotation coppice; dc = direct combustion; cf = co-firing; btl = biomass to liquid (2nd gen. biofuels); AN = Annuitiy; IRR = Internal Rate of Return; NPV = Net Present Value; AGM = Annual Gross Margin; RLH = Repetitive Latin Hypercube Sampling; MCS = Monte Carlo Simulation; EU = Expected Utility Approach; DU = Discounted Utility; SA = Sensitivity Analysis; BKT = Boosted Regression Tree; LCP = Linear Complementary Problem; CBA = Cost-Benefit Analysis, CA = Cost Accounting, and BEC = Break-Even Cost Analysis

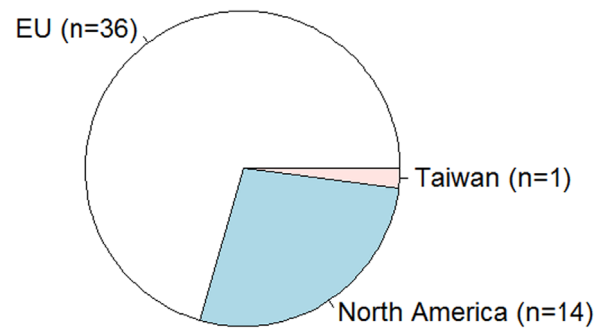


Fig. 2. Published papers on Miscanthus economics differentiated by regions. Total number of reviewed paper: $n = 51$.

reviewed literature together with selected core information extracted from the examined papers is provided in Table 1.

3.2. Literature description

3.2.1. Origin and date of publication

The 51 articles identified assess the situation from 16 different countries located in the EU⁶ (36 articles), North America (14 articles) and Asia (1 article) which might reflect the concentration of research regarding the production of this kind of biomass in some parts of the world (Fig. 2).

Within the EU, the UK ($n = 12$) and Ireland ($n = 11$) seem to be most involved in exploring the economics of Miscanthus cultivation. North America is mainly presented by the US with 12 articles. The remaining 16 papers are spread over 10 different countries. This concentration of Miscanthus research in very few countries also reveals the incentives set by individual national politics to foster the adoption of perennial biomass plants. For instance, the UK and Ireland are responsible for almost 2/3 of the articles published on EU countries. Both, the UK and Ireland, have in common, that they offer⁷ substantial establishment grants of up to 50% of the total costs of establishing src or Miscanthus plantations to farmers plus additional per ha payments of up to 80€ per year [9,45,47]. A similar promotion of perennials can be observed in the US under the “Biomass Crop Assistance Program” (BCAP) which offers establishment grants of up to 75% and support payments per ton or per ha of produced biomass [16,37].

The distribution of the appearance of papers (year of publication) over time is provided in Fig. 3, showing that the earliest identified paper originates from 1999.

An increase in publication activity is particularly revealed around the years 2008 and 2010, which might indicate an increased societal and political interest in economics of renewable energies. Again, specific policy measures could be a reason for the increased number of publications. The “Food, Conservation, and Energy Act” released in June 2008 for example, a US subsidization program, offering 288 billion US\$ over a period of five years to the agricultural sector including research and development of advanced biofuels and the production of biomass [74] could have incentivized research in this area. Another example for politics that is suitable to increase interest in biomass related research is a renewable energy feed in tariff (REFIT), released by the Irish government in 2007 to achieve, beside other targets, a 30% substitution of peat by biomass in selected power stations until 2015 [43,56]. However, this should not be overestimated since the total number of studies dealing with economics of Miscanthus remains

⁶ As no European studies outside of the EU have been found, EU and Europe are used interchangeably in our analysis.

⁷ For instance, “Energy Crop Scheme” (ECS) in the case of the UK [45] and the “Bioenergy Scheme” in Ireland [9].

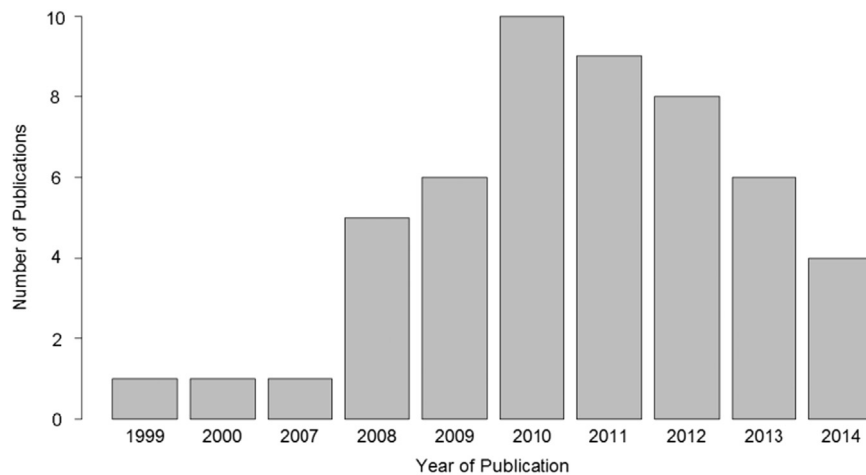


Fig. 3. Number of studies published per year between 1999 and 2014. The total number of papers ($n=51$) distributed according to their publishing year.

low, compared to other sources of renewable energy (see e.g. [75,76,77] for reviews on other renewable energy sources such as src, waste biomass and corn, respectively).

3.2.2. System of comparison

In order to assess the economic viability of *Miscanthus* relative to alternative crops or energy plants, the majority of publications choose to compare the cultivation of *Miscanthus* with the cultivation of short rotation coppice (src). More specifically, 23 out of the 51 publications (45%) used src as a reference system for *Miscanthus*. 13 articles (25%) do not mention any reference crop. Of those 23 publications assessing the economics of src, most focused on the short rotation cultivation of willow (15), 6 publications chose src poplar, 4 do not specify the kind of src and 2 publications assess the economics of src eucalyptus.⁸ Note that due to the fact, that some publications use more than one src plant as reference crop, the numbers do not sum up to 23. The second largest share of reference crops is represented by large grasses and here especially Switchgrass plays a major role (see Table 1). Depending on the cropping site *Miscanthus* most often shows substantially higher yield potentials than Switchgrass [15,16] although there are also examples where Switchgrass outmatches *Miscanthus* in terms of per ha yields [32]. Only few articles compared the profitability of *Miscanthus* with “traditional” crops such as wheat or corn, which often are the most dominant form of land use. Another area for comparison of *Miscanthus* is with fossil fuels in 5 articles to assess the competitiveness of *Miscanthus* as potential substitute mainly for coal. Even more concentrated than the origin of publications is the focus on the potential utilization pathways of *Miscanthus*. Of those publications that consider at least one *Miscanthus* utilization explicitly ($n=41$), 100% regard *Miscanthus* as feedstock for heat and/or energy production. For the majority of publications ($n=34$) a direct use of *Miscanthus* as energy carrier either as separate combustion material or as feedstock for co-firing with coal or peat is the method of choice to generate value from *Miscanthus*. Although the potential to produce advanced biofuels based on ligno-cellulosic feedstock such as *Miscanthus* is mentioned in 21 articles this alternative utilization is mostly referred

to very superficially. For instance, [23] states that the production of 2nd generation biofuels made from ligno-cellulose has not yet been commercialized and thus currently does not encourage the production of perennial energy crops. Other papers (e.g. [52,71]) simply mention the potential suitability of *Miscanthus* for further processed products without further specification, details and quantitative analysis. A deeper examination of the economics of ligno-cellulosic bioethanol production in Ireland is conducted by Deverell et al. [55] and Spicer et al. [33], who partly come to differing results. While both are concluding that, under a reformed tax regime ligno-cellulosic biofuels could possibly become competitive with fossil fuels, Deverell et al. [55] show that with cheaper imports of bioethanol and biomass available, biofuel processors in Ireland would not be willing to demand domestically produced feedstock. Six publications [15,25,30,38,44,71] at least mention a possible use such as animal bedding or construction material and only two of them address the possibility of the generation of higher value products such as chemicals and plastics, again without further specification. With the aim of economic assessment, the reviewed literature strongly focuses on one-sided utilization of *Miscanthus*, enabling only a limited sales market for low value products (e.g. for energy generation), limiting also the attractiveness for cultivation of *Miscanthus* compared to other channels of use.

3.2.3. Methods of assessment applied in literature

At least 20 different methods were applied to assess the profitability of *Miscanthus*. The net present value (NPV) is the most frequently used measure of profitability and was used in 25 articles. Since the adoption of *Miscanthus* is an investment that has to generate returns over a certain period of time in order to cover the initial establishment costs, decision makers have to discount expected future cash flows, using an “appropriate” discount factor, to find out if the initial investment pays off. The NPV is one output value of the discounted cash flow method, showing today's value of future payments minus initial costs (other approaches based on discounted cash flows that are shown in Fig. 4 are the internal rate of return and the annuity method).

The detailed distribution of the different approaches is presented in Fig. 4. Although methods like surveys, expert interviews, geographic information tools (GIS) or choice experiments are no means of capital budgeting in the strict sense, they are nevertheless presented in Fig. 4 to provide a comprehensive overview of the applied methods.

Due to the heterogeneity of methodological approaches used, there is no consistency to be expected either with regard to the

⁸ Note that the focus on specific tree species can also be due to the fact that legal regulations on the use of species do exist. According to Aust et al. [78], EU member states are advised to list fast growing tree species that are suitable for src. For instance willow, poplar, hazel, silver birch, sycamore, sweet chestnut and lime are eligible species for the “Bioenergy Crop Scheme” in the UK (for more information see <https://www.gov.uk/energy-crops-scheme-terms-and-conditions-of-your-agreement>). In Germany, the corresponding species are willow, poplar, black locust, birch, ash-tree, alder and oak [78].

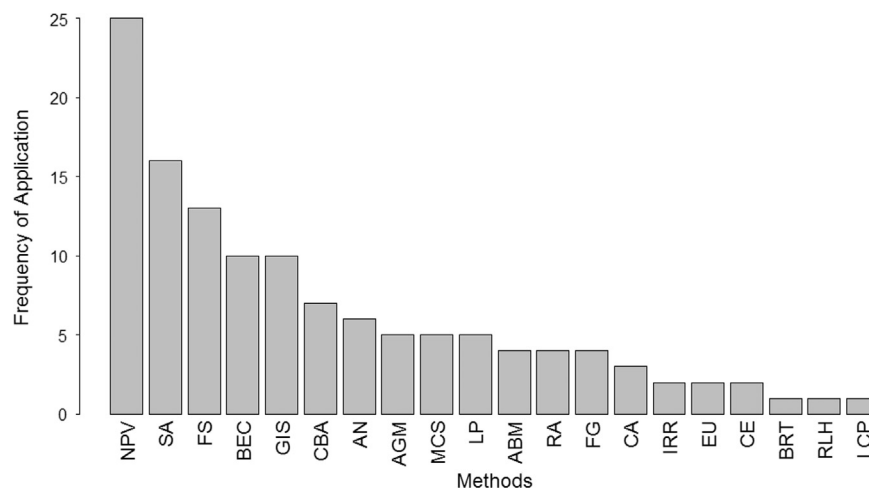


Fig. 4. Frequency of application of assessment criteria and research techniques used in literature. Explanation of abbreviations: NPV= Net Present Value, SA= Sensitivity Analysis, FS= Farm Survey, BEC= Break-Even Cost Analysis, GIS= Geographic Information System, CBA= Cost-Benefit Analysis, AN= Annuity, AGM= Annual Gross Margin, MCS= Monte Carlo Simulation, LP= Linear Programming Model, ABM= Agent-Based Model, RA= Regression Analysis, FG= Focus Group, CA= Cost Accounting, IRR= Internal Rate of Return, EU= Expected Utility Approach, CE= Choice Experiment, BRT= Boosted Regression Tree, RLH= Repetitive Latin Hypercube Sampling, and LCP= Linear Complementary Problem. Note that as the application of one model or technique does not exclude the application further techniques, the number of applications shown in Fig. 4 exceeds the total number of examined papers.

underlying research questions and assumptions or with regard to the results. Nevertheless it becomes apparent across several methodological approaches that there are drivers and, above all, barriers that influence the decision of farmers whether to invest in growing Miscanthus or not. The influencing factors that were identified by the literature review are examined and discussed in the following section.

4. Factors influencing the adoption and profitability of Miscanthus

This section provides an overview on several aspects that influence the profitability and adoption of Miscanthus. It is divided into three main subsections that cover the technical and agronomic assumptions on Miscanthus cultivation as well as the costs and benefits of Miscanthus cultivation. Next to addressing the assumptions on average values used in the reviewed studies, a particular focus was also on the variability and uncertainty underlying the assumptions made. This is due to the fact that the few studies on Miscanthus profitability that do actually account for risks find that the major barrier to Miscanthus cultivation is, besides eventual technical problems, the uncertainty that farmers are exposed to concerning the profitability of their investment in that crop [42,70]. Note that very mixed evidence was given by the studies whether Miscanthus does actually pay or not. Often, the reviewed studies refer to conditions (e.g. concerning future price and cost developments [37,43–45,62] or the availability of subsidies [40], contracts [23], suitable soils [37], etc.) under which Miscanthus adoption could be profitable. To a large extent, the still limited uptake by farmers in practice does reveal that some significant adoption barriers are present under current conditions in many countries. Partly, we are going to address the barriers of adoption in the here presented analysis.

4.1. Assumptions on Miscanthus cultivation

4.1.1. Lifespan

The total lifespan of Miscanthus is the period from establishment till closing of the Miscanthus plantation. As Miscanthus is a perennial crop, this lifespan differs greatly from those of traditional crops which are cultivated from year to year providing the

farmer with the opportunity to reallocate the land frequently to different crops. The long commitment of land to one single plant does not only limit the entrepreneurial freedom of the farmer which is seen as a major barrier to Miscanthus adoption by 7 of the reviewed papers [23,42,48,54,56–58]. It also implies the fact that Miscanthus needs a minimum period to break even with the initial investment costs. The assumptions made in the studies show a large variety and range between 10 years and 22 years [32,62,73,33] with a mean of 17 and a median of 16 years. Decreasing marginal biomass yields are claimed if exceeding the here indicated upper limits [33,79,80]. However, given these insights and given the fact that considering a longer lifetime and thus a larger number of harvest cycles can contribute to offset more of the initial costs, it is surprising that a 10 year lifecycle is assumed in some papers. The large variability in these assumptions reflects the high uncertainty with the production side of Miscanthus. This is due to the fact that Miscanthus is still a rather unexplored crop in many parts of the world and underlines the need for research and clear communication of agronomic insights in Miscanthus cultivation. Based on this background, a conservative assumption on the lifetime might be perceived by some authors as (misleading⁹) way to resolve problems of uncertainty. Fig. 5 shows the great discrepancy with respect to the underlying lifespan assumptions in literature. While the range of the assumed lifetime is largest for Continental Europe followed by North America, UK and Ireland show slightly more consistent assumptions concerning the lifespan of Miscanthus. This might be due to the higher diversity of regions, climates etc. and the resulting data in Continental Europe and North America, while UK and Ireland face more similar production conditions. In addition, research dealing with the economics of Miscanthus in the UK and Ireland is strongly concentrated to a few research groups which might result in a more uniform database than in other regions.

4.1.2. Establishment-to-revenue period

As indicated in Section 2, a period with strongly limited yields has to be overcome until the plantation is fully established and Miscanthus reaches its regular level of productivity [44]. During this

⁹ Conservative assumptions only would result in a critical level of undervaluing of any investment opportunity and thus result in strongly biased recommendations to farmers. A better methodological basis to deal with uncertainty is needed.

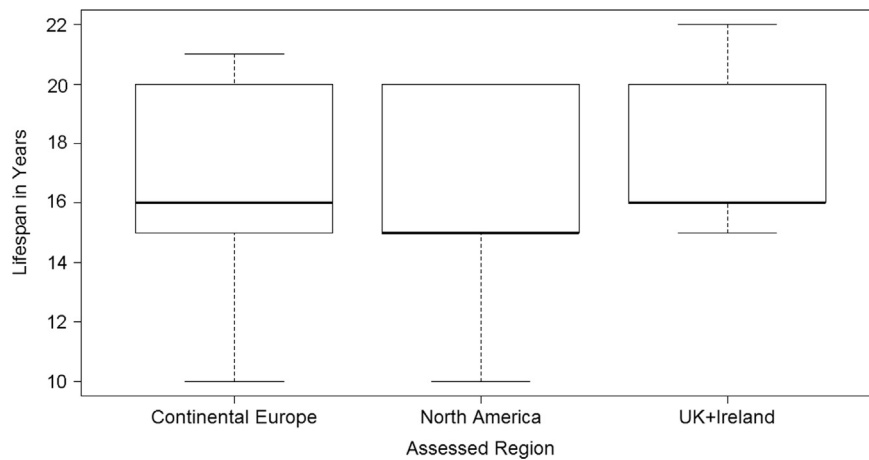


Fig. 5. Assumed *Miscanthus* lifespan differentiated by region. The individual countries are summarized to larger aggregates in Fig. 5 for improved comprehensibility. The number of considered studies is 10, 10 and 14 for Continental Europe, North America and UK+Ireland respectively. The thick lines inside the boxes represent the respective median value.

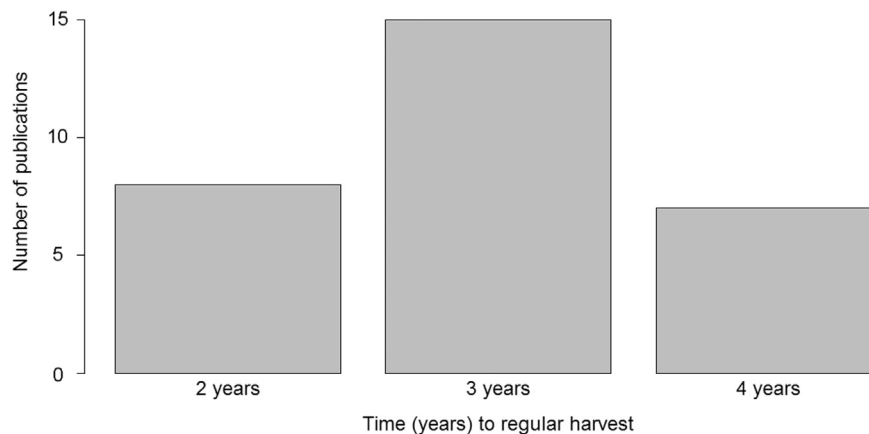


Fig. 6. Expectations with regard to the date of first harvest with regular yields. 30 out of the 51 studies report a first regular harvest year (2 years: $n=8$; 3 years: $n=15$; 4 years: $n=7$).

establishment period costs for e.g. planting [9,49] and maintenance [34,71] occur while no revenues are generated [41] which may result in liquidity shortages of the farm-household [67]. The assumed first harvest of a fully operational *Miscanthus* plantation that provides regular yields varies between years 2 and 4 [10,16] with a mode of year 3. Assuming that *Miscanthus* is planted in spring after last frosts (see Section 2), almost one year later (year 2; February till April [22]), *Miscanthus* shoots can be shredded and left on the field since harvest is not yet rational. One year later (year 3) first harvest can normally be conducted but with considerable limited yields due to the fact, that *Miscanthus* is still establishing. From year 4 onwards, (which means three years after planting), *Miscanthus* reaches its regular yield level. Considering this yield level development, the assumptions found in literature indicating regular yields from year 2 to 3 onwards might be too optimistic.

Fig. 6 illustrates the distribution of expectations concerning first harvest which are stated in the examined literature. The variability found in these assumptions, again underlines the great uncertainty associated with the agronomic factors underlying *Miscanthus* production and thus the need for increased field trials and communication.

4.2. Costs of *Miscanthus* cultivation

Growing *Miscanthus* is related to various specific costs which differ by nature and amount from the costs caused by

conventional annual crops. Those costs consist of single payments which are represented by the initial investment costs at the beginning and costs for closing the plantation at the end of its lifetime. Further, annual costs occur for harvest, transport and storage and maintenance. Additionally, opportunity costs of land and capital have to be taken into account. The former depending on potential alternative uses of the land committed to *Miscanthus*, the latter, accounted for by applying a certain discount rate, depending on potential revenues of alternative investments. Not all of these costs are of same relevance for the adoption decision of the farmer. In this section, these aspects are addressed in the given, i.e. chronological, order.

4.2.1. Non-recurring payments

4.2.1.1. Establishment of the plantation. The high upfront establishment costs are quoted as a major impediment for *Miscanthus* adoption [9,45,58,62]. However, the supply with detailed information about establishment cost is scarce in the examined literature. Therefore a comparison of the single cost items included in *Miscanthus* cultivation in all details is difficult. Thus, the following assessment is focused on the assumed total establishment costs of a *Miscanthus* plantation and the costs for *Miscanthus* rhizomes as they represent the largest share of total establishment costs. Fig. 7 shows the range of mean total establishment costs



Fig. 7. Mean total establishment costs and costs of rhizomes per ha. Total mean establishment costs include, but are not limited to, activities such as land preparation, planting and re-planting, plant protection and, as the largest share, costs for *Miscanthus* rhizomes.

quoted in literature ($n=19$) and the range of costs for *Miscanthus* rhizomes per ha ($n=12$)¹⁰.

The per ha costs for *Miscanthus* rhizomes range from 291.30€ to 11410.00€ [20,62] (both US). Note that these outliers are not displayed in Fig. 7. It has to be considered that the number of rhizomes used per ha varies strongly between 10,000 and 20,000 [36,61,9,49].¹¹ Assuming equal prices means a potential doubling of material costs. However prices for *Miscanthus* rhizomes found in literature also strongly vary between 0.03€ and 1.02€ [20,62] per piece. The “high cost scenario” for *Miscanthus* rhizomes assumed by [62] (1.02€/ per piece) relates to a field trial at the “Kellogg Biological Station” in 2008 (Michigan State University) and shows the extreme price differences for rhizomes in the US. It is never the less stated by James et al. [62] that, with rhizome prices decreasing to an European level through ongoing market maturity, cellulosic biomass from *Miscanthus* could become competitive with biomass from corn. In contrast, the very low price for rhizomes assumed by Khanna et al. [20] is based on data gained from Lewandowski et al. [22] where mechanical macro-propagation of rhizomes is assumed as a much cheaper alternative to the currently dominating in vitro micro-propagation of rhizomes. For further information about available propagation methods for *Miscanthus* rhizomes see e.g. [35,81]. Fig. 7 clearly shows that the costs for rhizomes represent the largest share of total establishment cost. When neglecting the above mentioned outliers, the mean per ha costs for rhizomes are 2074.95€ while the mean total establishment costs are 2575 €/ha. The average share of rhizome costs could therefore be estimated as around 80%. Thus, rhizome costs are the most crucial factor in *Miscanthus* establishment, but they should be reducible by further research for advanced propagation methods. Total establishment costs further include costs of labor and machinery required for planting activities. These costs range from 66 €/ha in a Canadian study [72] to 455 €/ha in a study for the UK [49] with a mean (over 4 studies) of 423 €/ha. This amount of costs corresponds to 16% of total establishment costs. Some studies that examine establishment costs assume that specific machinery, necessary to cultivate *Miscanthus*, is provided by a contractor so that costs caused by machinery are covered by the payment for the contractor [9,25,23] and no additional

machinery has to be purchased. The remaining 4% of costs are caused by establishment related maintenance such as weed control or fertilization and sum up to a mean of 112 €/ha. It is noteworthy that costs for replanting, although a replanting quota of up to 50% due to overwintering losses is indicated e.g. by Khanna et al. [20] and Clifton-Brown and Lewandowski [35], are usually not accounted for as establishment costs since they become due only in the second year or are even completely neglected.

4.2.1.2. Closing the plantation. Closing the plantation at the end of its lifetime also causes considerable costs between 77 €/ha [72] and 297 €/ha [10] with mean costs of 184.12 €/ha but as it only becomes necessary after 10–20 years, these costs are seen as less important from today’s perspective. For instance, assuming a discount rate of 5%, the 184 €/ha of costs for closing a plantation after $t=20$ years imply a present value of 69€. Thus, the costs of closing of the plantation is actually not relevant from an economic point of view, i.e. does not affect the decision taken by the farmer to adopt *Miscanthus*.

4.2.2. Annual costs

4.2.2.1. Harvest. Harvest costs play an important role since they account for the largest part of total annual costs. *Miscanthus* harvest can basically be carried out with standard farm equipment like corn harvesters and balers and is therefore assumed to be outsourced to a contractor as those machines are often too expensive to be maintained by single farms [25,9,23]. Costs for harvest vary due to different assumptions about processing methods such as chipping or mowing and baling [30,72] and because of an inconsistency in the definition and calculation of harvest cost. While in some studies harvesting includes transport and/ or storage [20,57,72] others only refer to the cost of cutting and baling or chipping [20,48,56]. Another barrier to comparability is the calculation of harvest costs whether on a per ha basis [20] or on a cost per weight basis [71]. When comparing harvest costs on a per ha basis it is important to account for the fact that the assumed per ha yield-level varies between the studies which may influence the assumed harvest cost. A specification of the average costs per unit of weight could therefore be helpful as well as an increased clarity concerning the components of the assumed harvest process. Fig. 8 displays the range of harvest cost. The mean harvest cost per ha is 270.50€. Over all studies that report harvest costs and without considering the above mentioned inconsistencies, per ha harvest costs range from 37.09 € (only fix harvest costs per ha without baling and field

¹⁰ Note that the latter are contained in the former in the presentation in Fig. 7.

¹¹ Note, however, that the assumption of 10000 rhizomes per ha is by far the most common one.

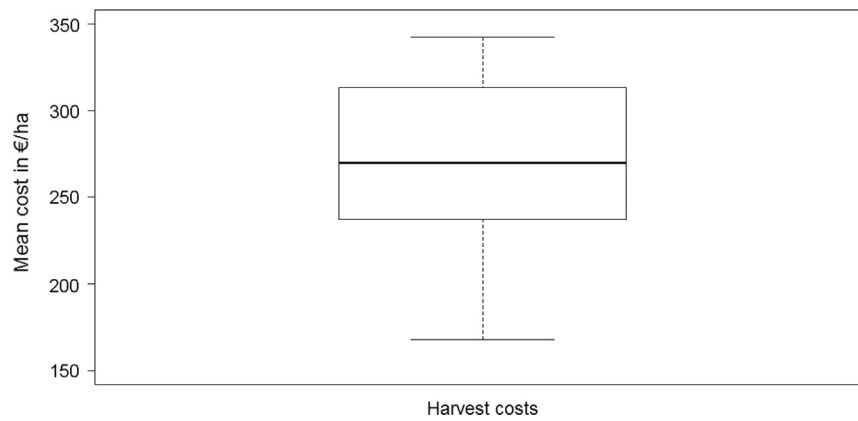


Fig. 8. Mean harvest cost per ha. Only those studies are included that provide harvest cost on a per ha basis ($n=17$).

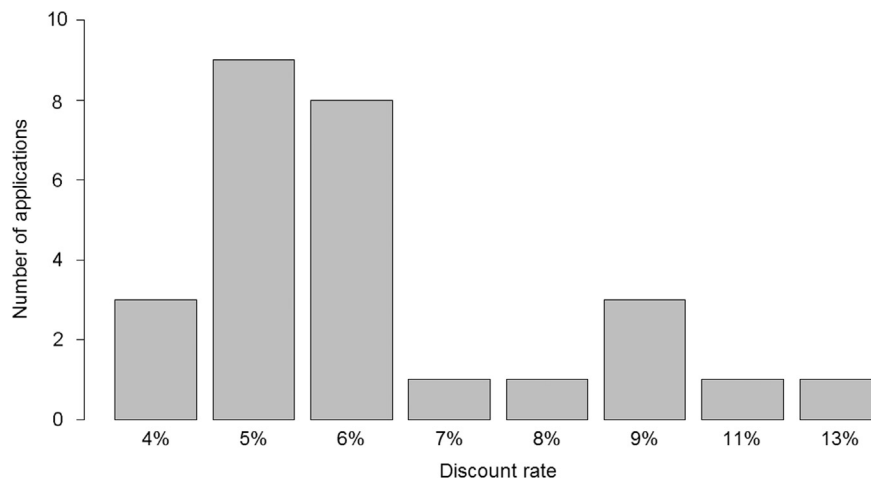


Fig. 9. Discount rates. Total number of discount rates found in literature is $n=28$. Note that Alexander et al. [52] states 5 different discount rates (6,8,9,11 and 13%) to consider e.g. future technology development. Other studies e.g. Clancy et al. [10] and Smeets et al. [25] do not report any discount rate with their results.

transport) [71] to 724.05€ (including mowing, conditioning, ranking, swathing, baling, staging, loading and storage) [20]. These outliers are not displayed in Fig. 8.

4.2.2.2. Transport and storage costs. Costs for transport and storage may affect farm profits severely [37,38,69] but they are difficult to define and compare as they are strongly influenced by external factors such as transport distances, infrastructure conditions, volume and weight which in turn depend on the harvest process, the moisture content, the time period of storage, fuel prices, etc. In addition, studies indicating costs for transport and storage are rare among the reviewed literature. Within those studies it is again problematic to compare costs for storage and transport on a per ha level [34] and on a per weight level [71]. Costs for transport are commonly stated as prices per oven dry ton (odt). They consist of a lump sum ranging from 0.99€ [37] to 5.76€ [48] per ton and an additional fee ranging from 0.06€ [15] to 0.36€ [49] per ton and km ($n=10$). Venturi et al. [30] provide an example on how the harvest process influences the costs for transport and storage: while chipping Miscanthus causes considerably lower harvest costs than baling, the subsequent transport costs are around 30% higher with 4.66 €/t plus 0.21 €/t and km compared to 3.69€ plus 0.16 €/t and km [30]. Additionally, due to the larger volume, storage costs are higher for chipped than for baled Miscanthus with 88 €/ha compared to 38 €/ha [30]. The mean of the assumed transport distance over all reviewed paper addressing this aspect is 49 km ($n=17$). Data on storage costs are rarely provided in the

studies examined ($n=8$) and range from 21 €/ha [33] to 95 €/ha [34] or from 3.97 €/t [71] to 6.79 €/t [34] also depending on factors like the harvest process as shown in the example. Further optimization of the production chain could help to reduce costs for transport and storage e.g. by establishing decentralized “down breaking” facilities as proposed by Trippe et al. [82,83] to achieve higher value density of the materials.

4.2.2.3. Fertilization, plant protection and weed control. According to Lewandowski et al. [22], weed control is only required in the establishment period and therefore limited to the first two years of cultivation. Moreover, only very few pests and diseases like Fusarium [84], Miscanthus Blight [85] and Barley Yellow Dwarf Luteovirus [86] are known to harm Miscanthus. In the examined literature Miscanthus is generally stated as low labor intensive plant with low nutrient requirements affording fertilization once a year after harvest [20]. Still, there is no consensus concerning nutrient requirements of Miscanthus, especially concerning the yield response to nitrogen applications [25]. The assumptions made by Khanna et al. [20] and Clifton-Brown and Lewandowski [87] to apply 60 kg N per ha and year can therefore only be seen as a benchmark. Other annual cost factors e.g. overhead costs are comparable to conventional crops and thus no particularity of Miscanthus production.

4.2.2.4. Opportunity costs of land and capital. Finally, imputed costs, the opportunity costs of land and capital are considered as

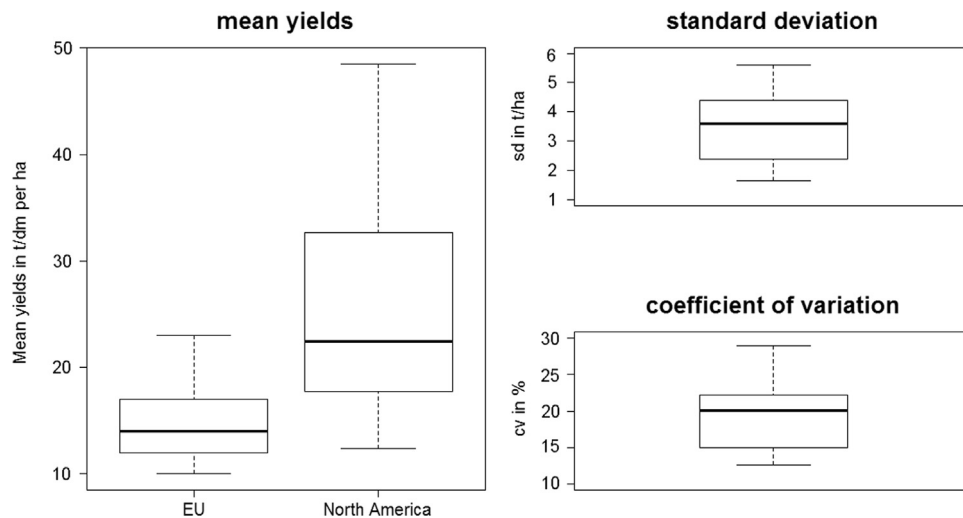


Fig. 10. Mean *Miscanthus* yields, standard deviations and coefficients of variation. Mean Yield: EU: $n=21$; North America: $n=13$; standard deviation: $n=12$; coefficient of variation (own calculation): $n=12$. Note that Aravindhakshan et al. [32] and Huang et al. [61] reported 2 and 3 assumptions on standard deviations respectively to account for different scenarios of uncertainty.

annual costs because the farmer, opting for *Miscanthus*, loses the opportunity to allocate land and capital to alternative uses each year again.

The costs of land are often not considered in literature [9,30,34] or assumed as the average regional land rent [25,34,66]. What is mentioned as a reason against the assumption that average land rents should be regarded as costs of land is, that land rents actually do not reflect the average price but the maximum willingness to pay to get an additional part of land. Land rents can therefore be considered as the marginal costs of land [23]. In most of the examined studies e.g. [15,20,23,51,72], the opportunity costs of land are considered as the “real” price of land because it is the loss of forgone revenues generated by alternative farm activities, that has to be overcome when opting for *Miscanthus*. Thus, the opportunity costs are mainly depending on the respective region and the potential alternatives to *Miscanthus*. Conclusively, when accounting for opportunity costs in a middle European setting, studies e.g. [44,45,9,48] compare *Miscanthus* with wheat, barley, rapeseed, sugar beet or potatoes while in southern Europe also crops like tomatoes or cotton may play a dominant role [66]. In contrast, corn, soybean and wheat production are the major farming businesses *Miscanthus* has to compete with in North America [16,71,62,37]. Hence it is obvious that the amount of opportunity costs cannot be stated in a general way. Growing *Miscanthus* on set aside land or on soils unsuitable for any other crop may cause very low opportunity costs while using highly productive farm land that could otherwise be used for special crops such as vegetables causes substantial opportunity costs. Insofar it is no surprise that only a minority ($n=8$; [15,20,54,60,66,70–72]) report opportunity costs explicitly, although 23 studies generally mention, that opportunity costs play an important role in the decision whether to grow *Miscanthus* or not. Of those studies reporting opportunity costs, 7 use alternative crops as a reference. In the US and Canada, the most profitable alternative to *Miscanthus* is a corn–soybean rotation, applied by Khanna et al. [20], Jain et al. [15], Vyn et al. [72], Shastri et al. [60] and Laporte et al. [71] to calculate opportunity costs ranging, for instance, from 71 €/t for corn to 177 €/t for soybeans [20]. For Germany [70], assumed opportunity costs range from 441 €/ha for winter barley to 558 €/ha for winter wheat.

Further, the opportunity costs of capital have to be accounted for, since the capital bound to *Miscanthus* cultivation cannot be invested otherwise to gain revenues. To represent farmer's

forgone revenues from alternative investment decisions as well as individual time preferences a discount rate has to be applied that reflects the time value of money.¹² In the reviewed literature the discount rates range from 4% [65] to 13% [52] with a mean of 5.8% and a median of 5% (Fig. 9).

Most frequently a discount rate of 5% ($n=9$) is applied in literature. According to James et al. [62] the average rate of return on capital for US farms between 1960 and 2001 was 5% leading to a corresponding discount rate of 5% to represent the average farmers opportunity cost of capital.

4.3. Benefits of *Miscanthus* cultivation

4.3.1. Yield assumptions

Similar to other arable crops, *Miscanthus* yield is one of the most decisive factors directly influencing farmers' income. In order to provide potential adopters with reliable information about expectable annual yields, validated and reliable assumptions are required. This applies even more in the case of a perennial crop that causes high upfront establishment costs. These costs have to be recovered over a long period of time leading to a loss of cropping flexibility which hinders the farmer to react quickly on changing market conditions [9]. The source of uncertainty connected to crop yields is the possible deviation of real annual yields from the expected potential yield. The higher this volatility is, the higher will be the uncertainty attached to *Miscanthus* yields. Given this assumption, it is remarkable that yield is considered as a stochastic factor when assessing the economics of *Miscanthus* only in 7 of the 51 studies [10,16,32,45,61,66,70]. If yield volatility is quantified, it is found to be large. For instance Alexander and Moran [45] and Clancy et al. [56] report standard deviations of 2.9 t/ha and 2.94 t/ha leading to coefficients of variation of 22.7% and 23% respectively. Across the reviewed literature the standard deviation ranges from 1.64 t/ha to 5.6 t/ha (both [61]) leading to corresponding coefficients of variation of 12.6–29%.¹³ Thus, also yield variability can be a decisive risk component in the cultivation of *Miscanthus*

¹² This may not only account for costs of borrowed capital, opportunity costs of equity capital but also for the riskiness of the investment decision (e.g. [18]).

¹³ Note that this relative variability of crop yields is high if compared with annual crops, where this production risk is known for a long time to be decisive for resource use decisions by farmers (e.g. [88,89]).

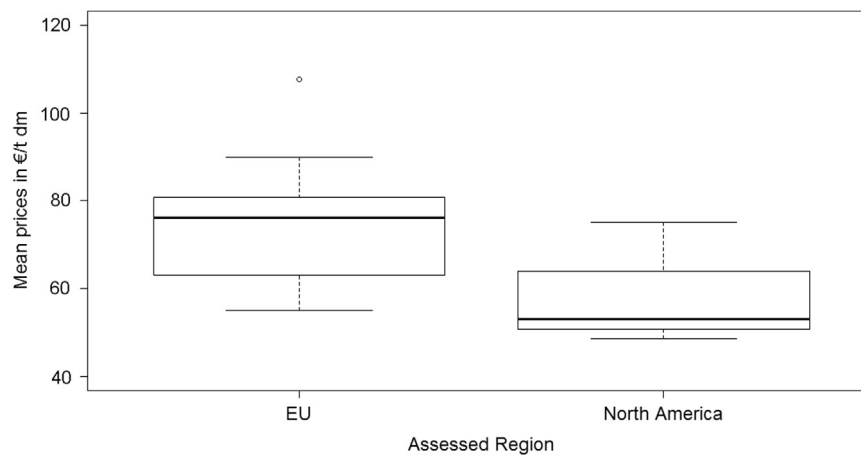


Fig. 11. Mean Miscanthus prices differentiated by region. EU: $n=18$; North America: $n=13$.

cultivation. Eight additional articles state that yield volatility could be expected but do not quantify this statement [15,30,38,40,41,67,71,72]. While the assessment of yield volatility by means of statistical measures falls short in the reviewed literature the number of studies reporting assumptions about minimum and maximum yield expectations is higher with $n=20$. Although assumptions on minimum and maximum yields give an impression of the potential range of yields, they may not be suitable as reliable yield indicators since they always represent extreme values that have to be considered in their respective context. For instance, Scheffran and BenDor [37] state a maximum expected yield level of 61 t dm/ha based on the assumption of high soil temperatures, high soil moisture and very few frost days in southern Illinois. That these optimal conditions are also very attractive to corn and soybean production making an adoption of Miscanthus in that area unlikely relativize those high Miscanthus yield assumptions. The range found in the examined studies between assumed minimum and maximum yields varies between 2.3 t/ha in the UK [51] and 41 t/ha in the US [15]. The wide range of yield expectations in the latter case results from the high diversity of examined planting sites concerning the type of soil, temperature or moisture etc. which again indicates the requirement of fertile locations for Miscanthus in order to generate highest yields. This is also supported by Alexander and Moran [45], while Dolginow et al. [16] states that growing Miscanthus on productive soils which are suitable for corn or soybeans is not a viable alternative due to high opportunity costs. Supporting the opinion of Dolginow et al. [16] studies, Jain et al. [15,69] and van der Hilst [40] recommend the adoption of Miscanthus on sites which are rather inappropriate for conventional crops e.g. due to eroded soils. The availability of arable land with low soil qualities may therefore be seen as factor driving farmers' decision towards the adoption of Miscanthus. This again shows the relevance of the trade-offs between high returns from Miscanthus cultivation and the opportunity costs associated with this planting decision in form of forgone revenues from other land use activities.

Fig. 10 summarizes the assumptions found in literature concerning mean Miscanthus yields differentiated by region, standard deviations and the corresponding coefficients of variation.

4.3.2. Prices

The single farmer is usually described as price taker which means he adjusts the production quantity to a market price which he cannot influence [90]. Two observations are common to many studies. First, there is no reliable "market price" for Miscanthus available as benchmark as there is only a very undeveloped market

for Miscanthus existing [48,54,60]. Second, current Miscanthus prices assumed in literature are often too low to make Miscanthus a competitive alternative to conventional crops [43–45]. Fig. 11 shows the broad range in average price assumptions made in the examined literature. A key problem is the difference of price relations used in literature. Prices were given relative to energy content or relative to a unit of weight and depending on the processing. Without information about the assumed moisture and energy content, problems arise in generating comparable data. To ensure consistent representation of prices we thus focus in Fig. 11 on prices that are related to weight in tons dry matter.

Price assumptions stated in the reviewed literature range from 48.56 €/t dm [15] to 134 €/t dm [52,36]. Assumptions on mean prices range from 48.56 €/t dm [15] to 108 €/t dm [52]. Reflecting volatilities on energy markets (e.g. the fossil fuel market), high price uncertainties should be expected in the case of Miscanthus. However, many studies do not consider price volatility when assessing the profitability of Miscanthus. James et al. [62] and Clancy et al. [56] for example are considering different price scenarios for Miscanthus in their models but only 7 out of the 51 examined studies assess the influence of price volatility on the profitability of Miscanthus by means of stochastic modeling. For instance, Clancy et al. [10] assume Miscanthus prices being normally distributed around a stochastic time trend and use them as input variables in the modified Monte Carlo simulation Repetitive Latin Hypercube sampling. A similar approach is conducted by Bocquého and Jacquet [67] where prices are supposed to follow a normal distribution. Due to a lack of historic data for energy crops, Bocquého and Jacquet [67] assume Miscanthus prices to show the same volatility as wheat prices. Thus, a coefficient of variation of 15% and 35% is applied as a measure of volatility in a low and a high price-volatility scenario assuming a mean Miscanthus output price of 70 €/t.

The general avoidance concerning the consideration of price uncertainty may be due to a lack of reliable price data which is caused by the relative novelty of the crop, considerable regional variation in yield potential and production costs and the lack of an established market that could otherwise offer easy access to information [16,72]. Thus, giving reliable estimations on Miscanthus prices is challenging. A solution for potential adopters to cope with price variability and uncertainty could be the production of Miscanthus based on contracts with e.g. energy suppliers. Binding prices and secure demand negotiated with contract partners is thus seen as a key factor, besides attractive establishment grants, to accelerate Miscanthus production by e.g. [45,43,48,52,68]. Overall, in 10 studies contracts offering guarantee prices leading to a risk sharing between producer and demander

of *Miscanthus* are seen as prerequisite for *Miscanthus* adoption [48,56, 67,23]. More research should therefore be undertaken concerning the optimal design and influences of contracts in *Miscanthus* production.

Instead of assuming a market price 11 studies [15,20, 32,36,55,61,62,65,71–73] calculate the breakeven price of *Miscanthus* production in order to identify the minimum required or the maximum achievable market price. For the calculation of the break-even price, two different approaches are applied in the examined studies: For instance, Jain et al. [15], Khanna et al. [20], and James et al. [61,62] calculate a “comparative breakeven price” which not only covers production costs of *Miscanthus* but also the forgone profits of alternative crops such as corn or soybean to find the minimum price that a farmer would demand for growing *Miscanthus*. Another approach can be found in studies where the competitiveness of *Miscanthus* relative to coal as a combustion material is assessed. Here, *Miscanthus* not only has to compete with other crops but also with fossil energy carriers. In this case not the breakeven price of *Miscanthus* necessary to cover all production costs is calculated but the price of *Miscanthus* which is required to break even with coal prices based on its respective energy content. This price is then compared to the production costs of *Miscanthus* with the result, that costs of biomass production can be competitive under certain circumstances [72] but are usually higher than its market value based on the energy content and the price of fossil energy carriers [20,32]. Thus, prices for fossil fuels are a major determinant of the profitability of *Miscanthus*. Since this review does not focus on utilization pathways of *Miscanthus* but on its cultivation, a large amount of literature dealing with the competitiveness of *Miscanthus* with regard to its calorific value has been excluded during the review process¹⁴ and is therefore not presented here. Break-even prices found in literature range from the minimum of 36 €/t [37,37] to a maximum of 284 €/t [36] with a mean of 76.11 €/t. The low break-even price calculated by Scheffran and BenDor [37] results from the fact that opportunity cost is not considered in this case. Taking into account opportunity costs of land, the break-even costs assumed by Scheffran and BenDor [37] rise to 51.21 €/t. When comparing the average mean “market” prices and the average mean breakeven prices it turns out that the mean *Miscanthus* break-even price is in average slightly higher with 76.11 €/t than the mean *Miscanthus* “market” price with 72.82 €/t. Although the underlying dataset cannot claim representativeness this could indicate that at current ratio of production costs to selling price production might not be profitable. However the difference between mean output price and mean breakeven price is only 3.30€ and is strongly influenced by the small data sample. Nevertheless, additional research and increased practical experience in *Miscanthus* cultivation could be helpful to lower the break-even costs which would make *Miscanthus* a more competitive source of biomass.

4.3.3. Subsidies

In 20 of the considered studies,¹⁵ policies to promote biomass production and/or utilization are regarded as requirement to adopt *Miscanthus* and/or to enhance the competitiveness of *Miscanthus* compared to alternative crops or resources. Establishment subsidies are seen as one of the most crucial preconditions for farmers to opt for *Miscanthus* cultivation. Some papers state that without the availability of substantial establishment grants,

farmers are unlikely to adopt *Miscanthus* facing the above mentioned high initial costs [44,45]. In the UK, the “Energy Crop Scheme” provides establishment grants of up to 50% or a certain fix amount of money to facilitate the adoption of *Miscanthus*. In detail, the subsidization ranges between a refund of 40–50% [52,50] of the establishment costs or, alternatively an establishment subsidy of 938–1072 €/ha [48,49]. A similar program is offered in Ireland, called “Bioenergy Scheme”, offering a refund of 50% of establishment costs or up to 1295–1450 €/ha [57,54]. Additionally there is an EU wide energy crop premium available of 45 €/ha per year [66,68,23]. In the US, the “Biomass Crop Assistance Program” (BCAP) offers support for biomass production but the considered implementation differs within the literature. Studies [60,37] assume that farmers are paid additional 40–45 €/t of *Miscanthus*, which only applies for a period of 2 years in the case of Shastri et al. [60]. In contrast, Dolginow et al. [16] assume that the BCAP offers an establishment grant of 75% of the initial costs plus additional 20 €/ha and year to support *Miscanthus* producers. Another form of government intervention to promote biomass for energy production is indirect subsidization, which serves to improve the competitiveness of biomass prices compared to fossil fuel prices. One of those measures demanded in literature e.g. by Aravindhakshan et al. [32] and Khanna et al. [20] is the taxation of carbon emissions caused by the combustion of fossil fuels. For example, [65] calculates a carbon tax of 15.8 €/t of CO₂ emitted by coal firing plants necessary to make *Miscanthus* competitive with coals for energy production. Other approaches for *Miscanthus* promotion mentioned in literature are mandatory co-firing with coals in power plants [32] and a monetization of ecosystem services provided by *Miscanthus* like sequestration of carbon in the soil, protection against erosion, waste water treatment, removal of heavy metals or increased biodiversity in order to reward the farmer for providing these public goods [20,23,24,25]. When comparing the area planted with *Miscanthus* in the UK with the area in other European countries like Germany, where no corresponding subsidies are available it seems as if subsidization of establishment costs is suitable to motivate farmers for *Miscanthus* adoption. According to Lorbacher [92] 10,000 ha are currently planted with *Miscanthus* in the UK while in Germany less than half the area (4500 ha) is committed to *Miscanthus* cultivation. There are certainly additional factors like socio-economic characteristics e.g. the age or the education of farmers that may also influence the farming decision [24,41], which makes it difficult to distinguish exactly between the effects caused by policy measures and those effects stimulated by other factors. Thus, the impact of subsidies on the farming decision concerning *Miscanthus* should be subject to further research. Especially the efficiency of subsidization measures with regard to partly contradicting policy targets like climate protection and energy supply security should be further assessed. Furthermore, it is important to mention that also the reduction of opportunity costs can contribute to the profitability of perennial bioenergy crops by reducing the extent to what these crops have to compete with other crops. For example, Lychnaras and Schneider [66] assessed the potential impact of the 2003 CAP reform (“decoupling” of agricultural payments, introduction of “cross-compliance”, etc. [93]) on the amount of biomass production. According to Lychnaras and Schneider [66], decoupled subsidies paid to farmers which are not devoted to a certain “conventional” crop lead to strongly decreased opportunity costs of biomass production. Under ceteris paribus conditions, these payments thus have a positive effect on the competitiveness of *Miscanthus* compared to other crops and concerning the adoption decision of farmers.

A more recent example for agricultural policies which could have an opportunity costs decreasing effect is the so-called “Greening” of the CAP, where farmers have to use 5% of their land

¹⁴ For more information about the calorific value of *Miscanthus* see e.g. [91]

¹⁵ Those are in detail: [16, 20, 23, 32, 33, 37, 38, 40, 44, 45, 47–51, 58, 59, 61, 64, 65].

as Ecological Focus Area (EFA) in order to receive some part of the direct payments [94]. While short rotation coppice can be accounted for as EFA (with a factor of 0.3), cropping *Miscanthus* however is not considered as potential contribution to these ecological focus areas. By that, *Miscanthus* is indirectly discriminated although research has already indicated that *Miscanthus* may contribute significantly to ecosystem services like increasing biodiversity¹⁶ compared to conventional crop rotations (through reduced application of agro-chemicals, reduced soil preparation and a harvest date in spring, offering cover for wildlife during winter), waste water treatment or carbon sequestration (e.g. [20,23,24,25]).¹⁷ To what extent these services (especially soil carbon sequestration and benefits for biodiversity) are relevant should be subject to further research. Adjusting current agricultural and environmental policies with regard to the benefits of *Miscanthus* would, c.p. increase the relative competitiveness of *Miscanthus* production.

5. Shortcomings in literature and implications for future research

A comprehensive assessment of the general profitability of *Miscanthus* is difficult due to the diverse nature of the underlying data sample gained from the reviewed literature. For instance, we found a large variety of different directions of research found in literature with the consequence of a high diversity in underlying assumptions and applied research methods. Considering *Miscanthus* cultivation as a business which is strongly affected by especially price and yield risks, leading to uncertain future cash-flows, it is surprising, that dynamic budgeting methods which do not account for uncertainty are preferred in the examined literature.¹⁸

In this context, the reluctance of authors with regard to the consideration of price and yield volatility is noticeable but can be assumed to result from a lack of historic yield and price data. To ameliorate the consideration of risk in future research dealing with the profitability of *Miscanthus*, budgeting methods taking into account uncertainty should be chosen instead of more simply applicable but unsophisticated dynamic approaches like discounted cash flow analysis. To enable further a more precise comparison of prices and costs across the literature, consistent measurements should be applied. Prices were found to be related to the energy content or to their mass measured in tons dry matter or in oven dry tons. A similar problem applies to costs of *Miscanthus* production which were found to be stated on a per hectare as well as on a per ton basis. A consistent measurement system including information about the assumed energy and moisture contents could be helpful to enhance the reliability of price and costs estimations. Additionally, across the examined studies, data are not reported accordingly thus leading to a lack of information in some parts e.g. regarding different cost items of cultivating *Miscanthus* such as costs of planting activities or costs of maintenance. While some papers are describing their assumed costs very precisely considering every single step of *Miscanthus*

cultivation like spraying, plowing, planting, etc. [20], others only report the total costs of all these activities together or in some cases they do not report any underlying assumption. Another example for incompleteness in the underlying assumptions can be found in a number of studies that analyse discounted future cash flows, but do not report the underlying discount rate. A similar problem can be found with regard to the reference system of *Miscanthus* where 25% of the examined papers do not mention any alternative crop to be compared with *Miscanthus* in order to assess its profitability. Further, in this context, 4 papers compare the profitability of growing *Miscanthus* with *src* but do not mention which kind of *src* cultivar is assumed, although considerable differences in the economics of *src* crops have been observed (e.g. [6]). Those lacks of transparency make it difficult to follow the calculations and conclusions stated in some papers. Since *Miscanthus* is still a rather novel and unexplored crop a lack of reliable and consistent data may need to be accepted, but also proves the necessity of intensive research for more trusted information about prices, yields, costs and cultivation of *Miscanthus*. It can also be assumed that further agronomic research, may be dealing with other breeds of *Miscanthus* than only *Miscanthus* × *Giganteus*, will lead to improved exploitation of currently unutilized potentials. Furthermore, a shortcoming identified in the examined studies is the strictly one-sided consideration of *Miscanthus* as a source of bioenergy while other potential applications which might allow higher sales prices and thus higher farm incomes remain completely untouched and should therefore be subject to future research.

6. Conclusion

In this article we assessed and reviewed information gained from 51 scientific studies in order to increase the knowledge concerning the economics of *Miscanthus* cultivation and to reveal and illustrate the factors influencing the adoption decision of farmers to provide future research and agricultural consultancy with key information and a solid data basis.

The factors that have been identified as being most critical for the profitability of *Miscanthus* and thus for the adoption decision of the farmer are diverse in their nature. More specifically, the expected lifespan, biomass yields, prices, establishment and opportunity costs and subsidization possibilities are crucial for the profitability of *Miscanthus* and therefore are seen as decisive factors for potential *Miscanthus* adopters.

Moreover, our review revealed large uncertainty especially concerning the key parameters yields and prices. Across all studies, mean yield assumptions were found to range from 10 to 48 t dm/ha while the assumptions concerning mean prices range from 48 to 134 €/t dm.

This indicates the need for further research in order to ensure more homogeneous assumptions on agronomic aspects (e.g. biomass yields, productive lifespan of *Miscanthus*, rhizome costs, etc.) as well as the application of more elaborate economic assessment methods, since static and even dynamic approaches which were mostly found in literature are not suitable to capture uncertainty involved in *Miscanthus* production. Our review revealed that, even though risks associated with the long-term commitment of *Miscanthus* production seem to be a major determinant for the adoption reluctance of farmers, only a limited number of studies addresses this explicitly, showing however that substantial variability in prices, yields and costs etc. can be expected.

As the assessment of bioenergy policies is not in the center of interest of this review, a deeper investigation of different national and transnational policies promoting different kinds of biomass could bring up more sophisticated results. Special emphasis

¹⁶ To measure beneficial effects of *Miscanthus* on biodiversity and soil carbon sequestration, the studies we reviewed compare *Miscanthus* to conventional, more intensive cultivation practices (see also Table 1 for the reference systems chosen in the reviewed studies). Compared to native, untouched nature or even semi-natural forests, biodiversity provided by *Miscanthus* plantations is low.

¹⁷ More general, any form of support for other (food or non-food) forms of land use *ceteris paribus* reduces the attractiveness of *Miscanthus*.

¹⁸ Along these lines, it is important to highlight that extension services addressing practitioners might even not fully take the step towards dynamic consideration of periodical cash-flows, but focus on the misleading perspective of static contribution margins (e.g. [95]).

should be placed on the optimal design of policies aiming at promoting the cultivation of *Miscanthus* and comparable perennial biomass crops. Further, as the absence of an established market for biomass is mentioned as major impediment leading to increased market risks, future research should focus on the potential to foster the adoption of *Miscanthus* by providing risk minimization to adopters e.g. by means of contracting and/or insurance solutions.

The great heterogeneity in assumptions, regarded regions, reference systems etc. do not allow a final and general statement concerning the profitability of *Miscanthus* as it always depends on individual aspects such as regional demand, the availability of subsidies or opportunity costs. Nevertheless this review paper provides a valuable overview about the most crucial factors influencing *Miscanthus* adoption which may help to encourage further research efforts and underlines the high but still unexploited potential of *Miscanthus* to contribute to future (energy) needs. Due to its relative novelty, *Miscanthus* and its cultivation is likely to be further developed through agronomical and (bio-) technological research, leading to improved yields and technical characteristics as well as reduced production costs [25].

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